



資料結構

Data Structure

Bonus 2

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Bonus02-Q1

Valid Sudoku

Code

```
class Solution {
public:
    bool isValidSudoku(vector<vector<char>>& board) {

        //用來記錄每一列是否出現過某個數字
        bool rows[9][9] = {};

        //用來記錄每一行是否出現過某個數字
        bool cols[9][9] = {};

        //用來記錄每個 3x3 宮格是否出現過某個數字
        bool boxes[9][9] = {};

        //逐格檢查整個數獨
        for (int i = 0; i < 9; i++) {

            for (int j = 0; j < 9; j++) {

                //如果目前位置是空格，直接跳過
                if (board[i][j] == '.')
                    continue;

                //將字元數字轉成陣列索引
                //例如 '1' -> 0、'9' -> 8
                int num = board[i][j] - '1';

                //計算目前數字位於哪個 3x3 宮格
                //宮格編號範圍為 0 ~ 8
                int boxIndex = (i / 3) * 3 + (j / 3);

                //如果該數字已經在同一列、同一行
                //或同一個宮格出現過
                //代表不符合數獨規則
                if (rows[i][num] ||
```

```

        cols[j][num] ||
        boxes[boxIndex][num])) {

            return false;
        }

        //紀錄這個數字已經出現過
        rows[i][num] = true;
        cols[j][num] = true;
        boxes[boxIndex][num] = true;
    }
}

//全部檢查完都沒有重複
//代表這個數獨盤面有效
return true;
}
};

```

Result

The screenshot displays the LeetCode interface for the 'Valid Sudoku' problem. On the left, there's a bar chart showing the runtime performance of various solutions, with the top solution using 14 ms and 0.99% of solutions. Below the chart, the C++ code for the solution is shown. The code defines a class Solution with a method isValidSudoku that checks if a 9x9 grid is a valid Sudoku. The method uses three 2D arrays (rows, cols, boxes) to track the numbers already present in each row, column, and 3x3 box. The test results on the right show that the solution is 'Accepted' with a runtime of 0 ms.

```

1 class Solution {
2 public:
3     bool isValidSudoku(vector<vector<char>>& board) {
4         // 用 bitset 或 bool 陣列記錄每行、每列、每個 3x3 宮格中出現過的數字
5         bool rows[9][9] = {};
6         bool cols[9][9] = {};
7         bool boxes[9][9] = {};
8     }
9 };

```